

Arjuna JEE 2025

Mathematics

Straight Lines

DPP: 2

Q1 The co-ordinates of the middle points of the sides of a triangle are $(4, 2)$, $(3, 3)$ and $(2, 2)$, then the co-ordinates of centroid are

- (A) $(3, 7/3)$
 (B) $(3, 3)$
 (C) $(4, 3)$
 (D) $(3, 4)$

Q2 If α, β and γ are the roots of equation $x^3 - 12x^2 + 44x - 48 = 0$. The centroid of the Δ whose co-ordinates are

$A(\alpha, \frac{1}{\alpha})$, $B(\beta, \frac{1}{\beta})$ and $C(\gamma, \frac{1}{\gamma})$, will be

- (A) $(4, \frac{11}{36})$
 (B) $(3, -1)$
 (C) $(-3, 5)$
 (D) $(-3, 1)$

Q3 The orthocentre of the triangle formed by the lines $xy = 0$ and $x + y = 1$ is

- (A) $(\frac{1}{2}, \frac{1}{2})$
 (B) $(\frac{1}{3}, \frac{1}{3})$
 (C) $(0, 0)$
 (D) $(\frac{1}{4}, \frac{1}{4})$

Q4 The vertices of a triangle are $A(0, 0)$, $B(0, 2)$ and $C(2, 0)$. The distance between the circumcentre and orthocentre is

- (A) 0
 (B) $\sqrt{2}$
 (C) $1/\sqrt{2}$
 (D) None of these

Q5

The incentre of the triangle formed by the axes and the line $\frac{x}{a} + \frac{y}{b} = 1$ is -

- (A) $(\frac{a}{2}, \frac{b}{2})$
 (B) $(\frac{ab}{a+b+\sqrt{ab}}, \frac{ab}{a+b+\sqrt{ab}})$
 (C) $(\frac{a}{3}, \frac{b}{3})$
 (D) $(\frac{ab}{a+b+\sqrt{a^2+b^2}}, \frac{ab}{a+b+\sqrt{a^2+b^2}})$

Q6 Find the coordinates of incentre of the triangle formed by $3x - 4y = 17$; $y = 4$ and $12x + 5y = 12$ is

- (A) $(\frac{29}{27}, \frac{38}{9})$
 (B) $(\frac{29}{9}, \frac{38}{27})$
 (C) $(\frac{29}{9}, \frac{38}{9})$
 (D) $(\frac{29}{27}, \frac{38}{27})$

Q7 In ΔABC , if the orthocenter is $(1, 2)$ and the circumcenter is $(0, 0)$, then centroid of ΔABC is

- (A) $(1/2, 2/3)$
 (B) $(1/3, 2/3)$
 (C) $(2/3, 1)$
 (D) none of these

Q8 If $A(2, 3)$, $B(3, 1)$ and $C(5, 3)$ are three points, then the slope of the line passing through A and bisecting BC is-

- (A) $\frac{1}{2}$
 (B) -2
 (C) $-\frac{1}{2}$
 (D) 2

Q9



Find the equation to the locus of a point which moves so that

Its distance from the axis of x is three times its distance from the axis of y .

Q10 Find the equation to the locus of a point which moves so that Its distance from the point $(a, 0)$ is always four times its distance from the axis of y .

Q11 Find the equation to the locus of a point which moves so that The sum of the squares of its distances from the axis is equal to 3 .

Q12 Find the equation to the locus of a point which moves so that The square of its distance from the point $(0, 2)$ is equal to 4.

Q13 Find the equation to the locus of a point which moves so that Its distance from the point $(3, 0)$ is three times its distance from $(0, 2)$.

Q14 Find the equation to the locus of a point which moves so that Its distance from the axis of x is always one half its distance from the origin.

Q15 Locus of the centroid of triangle whose vertices are $(a \cos t, a \sin t)$, $(b \sin t, -b \cos t)$ and $(1, 0)$, where t is a parameter, is

(A) $(3x - 1)^2 + (3y)^2 = a^2 - b^2$

(B) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$

(C) $(3x + 1)^2 + (3y)^2 = a^2 + b^2$

(D) $(3x + 1)^2 + (3y)^2 = a^2 - b^2$

Q16 Find the locus of a point whose distance from the point $(1, 2)$ is equal to its distance from the axis of y .

Q17 Find the locus of a movable point P , where its distance from the origin is 3 time its distance from y -axis.

Q18

Find the locus of a movable point P , for which the sum of its distance from $(0, 3)$ and $(0, -3)$ is 8 .

Q19 If the co-ordinates of a variable point P be $(at^2, 2at)$, find the locus of P

Q20 Find the locus of a movable point P , for which the difference of its distances from $(2, 0)$ and $(-2, 0)$ is 6 .

Q21 If P be the mid-point of the straight line joining the points $A(1, 2)$ and Q where Q is a variable point on the curve $x^2 + y^2 + x + y = 0$. Find the locus of P .

Q22 A stick of length l slides with its ends on two mutually perpendicular lines. Find the locus of the mid-point of the stick.

Q23 If $A(1, 1)$ and $B(-2, 3)$ are two fixed points, find the locus of a point P so that the area of $\triangle PAB$ is 9 sq.u.

Q24 A variable line cuts the x and y axes at A and B respectively where $OA = a$, $OB = b$ (O the origin) such that $a^2 + b^2 = 27$, find the locus of the centroid of $\triangle OAB$



Answer Key

Q1 (A)

Q2 (A)

Q3 (C)

Q4 (B)

Q5 (D)

Q6 (B)

Q7 (B)

Q8 (C)

Q9 $3x=y$

Q10 $15x^2 - y^2 + 2ax - a^2 = 0$

Q11 $x^2 + y^2 = 3$

Q12 $x^2 + y^2 - 4y = 0$

Q13 $8x^2 + 8y^2 - 36y + 6x + 27 = 0$

Q14 $3y^2 - x^2 = 0$

Q15 (B)

Q16 $y^2 - 2x - 4y + 5 = 0$

Q17 $3x^2 - y^2 = 0$

Q18 $7k^2 + 16h^2 - 112 = 0$

Q19 $y^2 = 4ax$

Q20 $5x^2 + 9y^2 = 45$

Q21 $2x^2 + 2y^2 - x - 3y + 1 = 0$

Q22 $x^2 + y^2 = l^2/4$

Q23 $2x + 3y + 13 = 0$

Q24 $x^2 + y^2 = 3$



[Android App](#) | [iOS App](#) | [PW Website](#)